## Midterm: what to know

**Format.** In class, 80 minutes, closed book, no notes, no calculators. Three problems, one per page, 100 points. Covers Lectures 1–11 (Lectures 1–2, 3–4, 5–8, 9–11 in the notes).

**The three problems.**

(i) **State, prove, apply.** One of the pre-registered theorems in list C below, stated exactly, proved from scratch, then used in a computation that also asks for a definition from list A.

(ii) **State and prove.** A second theorem from list C. Its full statement is printed on the exam; you prove it.

(iii) **Define, state, compute.** Definitions from list A, stated exactly; the relevant theorems from list B are printed on the exam; then a computation with them.

When proving a pre-registered theorem you may quote any other result from the notes that does not depend on the one you are proving; say what you are quoting. Short answers are expected: a correct argument of five lines beats a page.

**A. Definitions to know.** Be able to state each exactly, in the form the notes give it.

- The characteristic function of a random vector, and a Gaussian vector $\mathcal{N}(\mu, \Sigma)$ defined through it (Lectures 1–2, Definitions 1.2–1.3). Know why the definition allows singular $\Sigma$.
- Conditional expectation $\mathbb{E}[X \mid Y]$ as the orthogonal projection onto functions of $Y$, and its characterization as the $L^2$-best predictor (Lectures 1–2, Definition 2.2, Lemma 2.3).
- Convergence in distribution (Lectures 3–4, Definition 1.1).
- A triangular array, $s_n^2$, the normalized row $Y_{n,i} = X_{n,i}/s_n$, and Lindeberg's condition $L_n(\varepsilon) \to 0$ (Lectures 3–4, Definition 1.8).
- A pair partition of $\{1, \dots, 2q\}$, and the number $(2q-1)!!$ of them (Lectures 5–8, Definition 1.1).
- The GOE matrix ("Gaussian Wigner matrix" in the notes): $M = \frac{1}{\sqrt{2N}}(A + A^\top)$ with $A$ i.i.d. $\mathcal{N}(0,1)$; equivalently symmetric with independent entries, $\operatorname{Var} M_{ij} = \frac{1}{N}$ off the diagonal and $\frac{2}{N}$ on it; and its covariance $\mathbb{E}[M_{ij}M_{kl}] = \frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ (Lectures 5–8, Definition 1.7).
- The resolvent and the Stieltjes transform $m_N(z) = \frac{1}{N}\operatorname{tr}(M - zI)^{-1} = \frac{1}{N}\sum_i \frac{1}{\lambda_i - z}$ (Lectures 5–8, Definition 3.1).
- The semicircle density $\rho_{sc}(x) = \frac{1}{2\pi}\sqrt{4 - x^2}$ on $[-2, 2]$ (Lectures 5–8, Theorem 2.1).
- A Gaussian process, its mean function $\mu_t$ and covariance kernel $\Sigma_{s,t}$; a centered process (Lectures 9–11, Definition 1.1).
- Brownian motion on $[0, 1]$: the centered Gaussian process with kernel $\Sigma_{s,t} = \min(s, t)$ (Lectures 9–11, Example 1.6).
- The canonical metric $d(s, t) = (\mathbb{E}(X_s - X_t)^2)^{1/2}$ (Lectures 9–11, Definition 1.8).
- An $\varepsilon$-net, an $\varepsilon$-separated set, and the covering number $N(T, d, \varepsilon)$ of a metric space (Lectures 9–11, Definition 2.5 and Lemma 2.6).
- The Gaussian width $\mathrm{w}(S) = \mathbb{E}\sup_{h \in S}\langle h, g\rangle$ of a set $S \subset \mathbb{R}^k$ (Lectures 9–11, Exercise 1.14).

**B. Theorem statements to know.** Be able to state each exactly and to use it; the proof is not examined. On the exam, any of these may be printed for you to use, or you may be asked to state one.

- Characteristic functions determine the law; a joint characteristic function that factorizes means independence (Lectures 1–2, Proposition 1.1).
- The parameters of $\mathcal{N}(\mu, \Sigma)$ are the mean and covariance; one coordinate at a time: $X$ is Gaussian iff every $\langle a, X\rangle$ is (Lectures 1–2, Proposition 1.4, Corollary 1.7).
- Uncorrelated jointly Gaussian variables are independent (Lectures 1–2, Theorem 1.16).
- The rules of conditional expectation: $\mathbb{E}[g(Y)X \mid Y] = g(Y)\mathbb{E}[X \mid Y]$; $\mathbb{E}[X \mid Y] = \mathbb{E}X$ when $X$ is independent of $Y$; the tower property; the law of total variance (Lectures 1–2, Proposition 2.4, Corollary 2.5).
- The conditional law $X \mid Y \sim \mathcal{N}(AY, \Sigma_{XX} - A\Sigma_{YX})$ with $A = \Sigma_{XY}\Sigma_{YY}^{-1}$ (Lectures 1–2, Theorem 2.10).
- Convergence in distribution follows from convergence of $\mathbb{E}f$ for smooth $f$ (Lectures 3–4, Lemma 1.2); Lindeberg's theorem (Theorem 1.9).
- Wick's theorem, and its consequences: odd moments vanish; $\mathbb{E}g^{2q} = (2q-1)!!$; Gaussian integration by parts $\mathbb{E}[X_a F(X)] = \sum_j \Sigma_{aj}\mathbb{E}[\partial_j F(X)]$ (Lectures 5–8, Theorem 1.2 and §1.2).
- Wigner's semicircle law, in both forms: convergence of $\frac{1}{N}\operatorname{tr} M^k$ to the moments $\int x^k \rho_{sc}$ (Lectures 5–8, Theorem 2.1), and convergence of the proportion of eigenvalues in an interval to $\int_a^b \rho_{sc}$ (Theorem 4.19). The even moments of $\rho_{sc}$ are the Catalan numbers, $C_1, \dots, C_4 = 1, 2, 5, 14$ (Propositions 2.16–2.17).
- The inversion formula for the count of eigenvalues in $(a, b)$, and the local-count bound $\frac{1}{N}\#\{|\lambda_i - a| \le \eta\} \le 2\eta\, \Im m_N(a + i\eta)$ (Lectures 5–8, Theorem 3.4, Lemma 3.6).
- The union bound for the maximum of $n$ centered Gaussians, in expectation and in tail form, with no independence assumed; the independent case attains $\sqrt{2\log n}$; the one-scale bound $D\sqrt{2\log n}$ (Lectures 9–11, Propositions 1.9, 2.2, 2.3, Corollary 2.4).
- Increments of a Gaussian process are Gaussian with standard deviation $d(s, t)$ (Lectures 9–11, Lemma 2.1).
- Covering numbers of the sphere: $N(S^{d-1}, \|\cdot\|, \varepsilon) \le (1 + 2/\varepsilon)^d$ and $N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \ge 2^{d-2}$ (Lectures 9–11, Lemma 2.6, Exercise 2.15).
- Dudley's inequality (Lectures 9–11, Theorem 2.9); Sudakov's inequality (the quoted box in §2.6).

**C. Theorems to know with proof — the pre-registered list.** Two of the following will be on the exam, stated as here, for you to prove from scratch. Prepare all of them.

**T1.** *Affine images* (Lectures 1–2, Theorem 1.5). If $X \sim \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^d$ and $A \in \mathbb{R}^{m\times d}$, $b \in \mathbb{R}^m$, then $AX + b \sim \mathcal{N}(A\mu + b,\ A\Sigma A^\top)$.

**T2.** *Conditioning as projection* (Lectures 1–2, Theorems 2.9 and 2.10). Let $(X, Y)$ be jointly Gaussian and centered with $\Sigma_{YY}$ invertible, and put $A = \Sigma_{XY}\Sigma_{YY}^{-1}$. Then $X - AY$ is independent of $Y$, $\mathbb{E}[X \mid Y] = AY$, and $X \mid Y \sim \mathcal{N}\big(AY,\ \Sigma_{XX} - A\Sigma_{YX}\big)$. (You may quote Theorem 1.16: jointly Gaussian and uncorrelated implies independent.)

**T3.** *The central limit theorem by swapping* (Lectures 3–4, Theorem 1.7). Let $X_1, X_2, \dots$ be i.i.d. with mean 0, variance 1 and $\mathbb{E}|X_1|^3 < \infty$, let $S_n = n^{-1/2}\sum_{i\le n} X_i$ and $g \sim \mathcal{N}(0,1)$. For every $f \in C^3(\mathbb{R})$ with bounded third derivative,

$$\big|\mathbb{E}f(S_n) - \mathbb{E}f(g)\big| \le \frac{\|f'''\|_\infty}{6\sqrt{n}}\big(\mathbb{E}|X_1|^3 + \mathbb{E}|g|^3\big).$$

**T4.** *Lindeberg's theorem* (Lectures 3–4, Theorem 1.9). Let $\{X_{n,i}\}_{i\le k_n}$ be a triangular array, independent within rows, with $\mathbb{E}X_{n,i} = 0$, $s_n^2 = \sum_i \operatorname{Var} X_{n,i} \to \infty$, satisfying Lindeberg's condition: $\frac{1}{s_n^2}\sum_i \mathbb{E}[X_{n,i}^2 \mathbf{1}\{|X_{n,i}| > \varepsilon s_n\}] \to 0$ for every $\varepsilon > 0$. Then $s_n^{-1}\sum_i X_{n,i} \Rightarrow \mathcal{N}(0,1)$. (You may take as given the two-sided Taylor remainder bound $|R(u)| \le \min(\frac{\|f'''\|_\infty}{6}|u|^3, \|f''\|_\infty u^2)$ and the reduction to $C^3$ test functions, Lemma 1.2.)

**T5.** *Stein's lemma* (Lectures 3–4, Theorem 2.1). A random variable $W$ is standard Gaussian if and only if $\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)]$ for all bounded $f \in C^1(\mathbb{R})$ with bounded derivative. (Both directions. For the converse you may use that a distribution is determined by its characteristic function.)

**T6.** *Wick's theorem* (Lectures 5–8, Theorem 1.2). For $X \sim \mathcal{N}(0,\Sigma)$ and indices $a_1, \dots, a_{2q}$,

$$\mathbb{E}\big[X_{a_1}\cdots X_{a_{2q}}\big] = \sum_{\text{pair partitions } P}\ \prod_{\{r,s\}\in P} \Sigma_{a_r a_s},$$

and odd moments vanish.

**T7.** *The Gaussian Poincaré inequality* (Lectures 5–8, Lemma 4.2). For $X \sim \mathcal{N}(0, I_n)$ and $f \in C^1(\mathbb{R}^n)$ with bounded gradient, $\operatorname{Var} f(X) \le \frac{\pi^2}{8}\,\mathbb{E}\|\nabla f(X)\|^2$.

**T8.** *The maximum of n Gaussians* (Lectures 9–11, Propositions 1.9 and 2.2). If $X_1, \dots, X_n$ are centered Gaussians, not necessarily independent, with $\operatorname{Var} X_i \le \sigma^2$, then $\mathbb{E}\max_i X_i \le \sigma\sqrt{2\log n}$ and $\mathbb{P}\big(\max_i X_i \ge \sigma\sqrt{2\log n} + \sigma u\big) \le e^{-u^2/2}$ for $u \ge 0$.

**T9.** *Covering the sphere* (Lectures 9–11, Lemma 2.6). $N(S^{d-1}, \|\cdot\|, \varepsilon) \le (1 + 2/\varepsilon)^d$ for every $\varepsilon > 0$.

**T10.** *Dudley's inequality, finite index set* (Lectures 9–11, Theorem 2.9). For a centered Gaussian process on a finite $(T, d)$ with diameter $D$,

$$\mathbb{E}\max_{t\in T} X_t \ \le\ 12\int_0^{D/2} \sqrt{\log N(T,d,\varepsilon)}\, d\varepsilon.$$

(You may take the nets $T_k$, the maps $\pi_k$ and the links lemma, Definition 2.7 and Lemma 2.8, as given.)

**What is not examined.** The starred sections of Lectures 1–4; Marchenko–Pastur; the proofs behind the quoted boxes (Moore–Aronszajn, Sudakov's inequality, Talagrand's theorem); anything from Lecture 11.

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A practice midterm with solutions is posted beside this sheet. It has the same structure and the same length as the real one.